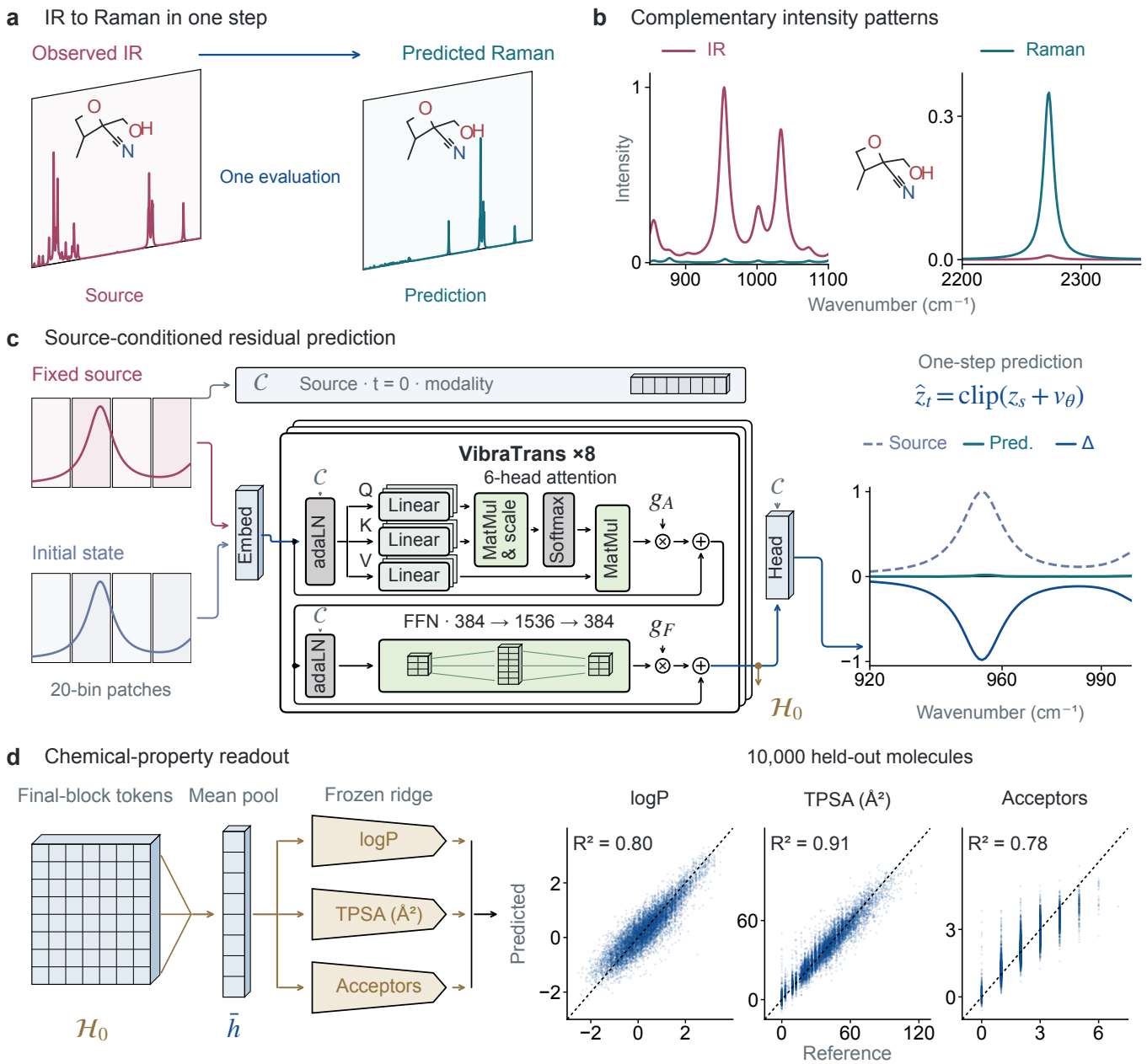

a
IR to Raman in one step
Observed IR
Predicted Raman
One evaluation
Source
Prediction
b
Complementary intensity patterns
IR
Raman
Intensity
Wavenumber (cm⁻¹)
c
Source-conditioned residual prediction
Fixed source
Initial state
20-bin patches
Source · t = 0 · modality
Embed
VibraTrans ×8
6-head attention
adaLN
Q
K
V
Linear
MatMul & scale
Softmax
MatMul
FFN · 384 → 1536 → 384
Head
One-step prediction
Source
Pred.
Δ
Wavenumber (cm⁻¹)
d
Chemical-property readout
Final-block tokens
Mean pool
Frozen ridge
logP
TPSA (Å²)
Acceptors
10,000 held-out molecules
R² = 0.80
R² = 0.91
R² = 0.78
Predicted
Reference